MA1503 Final Exam Tutorial

April 20, 2026

I. Short Answer and Proof Questions

1. Provide short answers to the following questions.

   (1) What is the definition of a **linear function** $f : \mathbb{R}^n \to \mathbb{R}^m$?

   (2) How many solutions can a linear system have?

   (3) What is the definition of a **subspace** of a vector space?

   (4) How can you compute the rank of an $m \times n$ matrix A?

   (5) How can you compute the inverse of an invertible $n \times n$ matrix A?

   (6) What region of $\mathbb{R}^2$ always has area equal to $\pm \det \begin{bmatrix} a & b \\ c & d \end{bmatrix}$? Draw and label a picture that represents this region.

   (7) The least-squares solutions to $Ax = b$ are the exact solutions to what matrix equation?

   (8) What $n \times n$ matrices have n orthonormal eigenvectors?

   (9) What is the definition of the **singular values** of a matrix A?

   (10) Suppose A is a $2 \times n$ matrix with a singular value decomposition $A = U\Sigma V^T$. Assume rank $A = 2$. Describe the shape $\{Av \in \mathbb{R}^2 : v \in \mathbb{R}^n \text{ with } \|v\| = 1\}$ and explain how this shape is related to the matrices U and Σ.

2. Let $A \in M_{n\times n}$. Suppose that A is invertible and $\lambda \in \mathbb{R}$ is an eigenvalue of A. Justify the following statements:

   (1) $\lambda \neq 0$.

   (2) λ^{-1} is an eigenvalue of A^{-1}.

3. Let $A \in M_{n\times n}$. Prove the following statements:

   (1) A and A^T have the same set of eigenvalues.

   (2) If λ is an eigenvalue of A, then $\lambda^2 + 1$ is an eigenvalue of $A^2 + I$.

4. Let $\mathbb{P}_3$ be the vector space of polynomials of degree at most 3 with real coefficients. Let

$$E_3 = \{p \in \mathbb{P}_3 : p(x) = p(-x) \text{ for every } x \in \mathbb{R}\}$$

   be the set of even polynomials in $\mathbb{P}_3$.

   (1) Give the definition of the statement: E_3 is a subspace of $\mathbb{P}_3$.

   (2) Show that E_3 is a subspace of $\mathbb{P}_3$.

(3) Find a basis of E_3. Justify your answer.

5. Consider the two matrices

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ 0 & a_{22} & a_{23} & a_{24} \\ 0 & 0 & a_{33} & a_{34} \end{bmatrix}, \quad B = \begin{bmatrix} b_{11} & b_{12} & b_{13} & b_{14} \\ 0 & b_{22} & b_{23} & b_{24} \\ 0 & 0 & 0 & b_{34} \end{bmatrix},$$

where $a_{11}, a_{22}, a_{33}, b_{11}, b_{22}$ and b_{34} are nonzero. Show that there does not exist a 3×3 invertible matrix C, such that $A = CB$.

II. Eigenvalue

6. Suppose A is a 3×3 matrix with all real entries, whose eigenvalues include the complex numbers 2 and $1 - i$.

   (1) Find a polynomial formula for the function $f(x) = \det(A^{-1} - xI)$.

   (2) Compute $f(5)$.

7. Suppose A is a 3×3 matrix that has exactly two distinct (complex) eigenvalues given by -1 and 2, and that has all three of the following vectors as eigenvectors:

$$\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}, \quad \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \quad \text{and} \quad \begin{bmatrix} 2 \\ -2 \\ 1 \end{bmatrix}$$

   (1) Can the matrix A be non-diagonalizable? If this is possible then give an example of such a matrix A, and otherwise explain why it is impossible.

   (2) Can the matrix A be non-invertible? If this is possible then give an example of such a matrix A, and otherwise explain why it is impossible.

   (3) Can the matrix A be orthogonal? (That is, can it hold that A is invertible with $A^{-1} = A^T$?) If this is possible then give an example of such a matrix A, and otherwise explain why it is impossible.

   (4) Can the matrix A be symmetric? (That is, can it hold that $A = A^T$?) If this is possible then give an example of such a matrix A, and otherwise explain why it is impossible.

III. Diagonalization

8. A is a 2×2 matrix and $-1 < \lambda < 1$ is a real number with

$$A \begin{bmatrix} 2 \\ 2 \end{bmatrix} = \lambda \begin{bmatrix} 2 \\ 2 \end{bmatrix} \quad \text{and} \quad A \begin{bmatrix} 2 \\ -2 \end{bmatrix} = \begin{bmatrix} 2 \\ -2 \end{bmatrix}.$$

Compute A and $\lim_{n \to \infty} A^n$. The entries in your answer for A should be expressions involving λ.

9. This problem has five parts.

   (1) Give an example of a diagonal square matrix that is not invertible.

   (2) Give an example of a diagonalizable square matrix that is not diagonal.

   (3) Give an example of a triangular square matrix that is not diagonalizable.

   (4) Give an example of an invertible square matrix that is not triangular.

   (5) Give an example of an orthogonal 2×2 matrix that is not a rotation matrix.

10. Let A be a symmetric 3×3 matrix with real entries. Suppose that its characteristic polynomial is $c_A(\lambda) = (1-\lambda)(2-\lambda)^2$ and that

$$Av = v, \quad \text{for } v = \begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix}.$$

   (1) Determine whether A is diagonalizable or not. Justify your answer.

   (2) Let $\lambda_1 < \lambda_2$ be eigenvalues of A. For each $j = 1, 2$, find the geometric multiplicity d_j of λ_j and a basis of the eigenspace $E_{\lambda_j}(A)$. Justify your answer carefully.

   (3) Find A. You may leave it as a product of matrices.

IV. Orthogonality

11. Suppose

$$v = \begin{bmatrix} 1 \\ 1 \\ -1 \\ 0 \end{bmatrix} \quad \text{and} \quad w = \begin{bmatrix} -1 \\ 1 \\ 1 \\ 1 \end{bmatrix}.$$

    Find a matrix A such that if $x \in \mathbb{R}^4$ then

$$Ax = (\text{ the vector in } \mathbb{R}\text{-span}\{v, w\} \text{ that is as close to } x \text{ as possible }).$$

12. Consider the plane $P = \left\{ \begin{bmatrix} x \\ y \\ z \end{bmatrix} \in \mathbb{R}^3 : 3x - y + 6z = 0 \right\}$ in $\mathbb{R}^3$.

    (1) The subspace P is 2-dimensional. Find an orthogonal basis for P.

    (2) Find the vector in P that is closest to $v = \begin{bmatrix} 3 \\ 2 \\ 1 \end{bmatrix}$.

V. Least Square

13. (1) Suppose $A = \begin{bmatrix} 1 & 3 \\ 0 & -1 \\ 2 & 2 \end{bmatrix}$ and $b = \begin{bmatrix} 2 \\ 1 \\ 2 \end{bmatrix}$. Does the equation $Ax = b$ have an exact solution? Find a solution or explain why none exists.

(2) Again suppose $A = \begin{bmatrix} 1 & 3 \\ 0 & -1 \\ 2 & 2 \end{bmatrix}$ and $b = \begin{bmatrix} 2 \\ 1 \\ 2 \end{bmatrix}$. Does the equation $Ax = b$ have a least-squares solution? Find a solution or explain why none exists.

(3) Suppose A is an $m \times n$ matrix and $b \in \mathbb{R}^m$. Indicate which of the following are TRUE or FALSE.

(a) If $x \in \mathbb{R}^n$ has $A^\top Ax = A^\top b$ then it always holds that $Ax = b$.

(b) If $x \in \mathbb{R}^n$ has $Ax = b$ then it always holds that $A^\top Ax = A^\top b$.

(c) If the equation $Ax = b$ has no solution then $A^\top Ax = A^\top b$ might also have no solution.

(d) If the equation $Ax = b$ has a unique solution then $A^\top Ax = A^\top b$ also has a unique solution.

(e) If the equation $A^\top Ax = A^\top b$ has a unique solution x then $Ax = b$ has at most one solution.

VI. Singular Value Decomposition

14. (1) Compute the singular values of the matrix $A = \begin{bmatrix} 1 & -1 \\ -2 & 2 \\ 2 & -2 \end{bmatrix}$.

(2) Suppose A is a 2×2 matrix with a singular value decomposition $A = U\Sigma V^T$ where U and V are orthogonal 2×2 matrices and $\Sigma = \begin{bmatrix} 10 & 0 \\ 0 & 5 \end{bmatrix}$. The first column of U is the vector $\begin{bmatrix} -4/5 \\ 3/5 \end{bmatrix}$. Draw a picture of the region in $\mathbb{R}^2$ given by

$$\left\{ Ax : x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \in \mathbb{R}^2 \text{ is a vector with } x_1^2 + x_2^2 \leq 1 \right\}.$$

Make your picture as detailed as possible.

(3) Find an orthonormal basis of $\mathbb{R}^3$ that contains the vector $\begin{bmatrix} 1/3 \\ -2/3 \\ 2/3 \end{bmatrix}$.

15. (1) Compute the singular values of the matrix $A = \begin{bmatrix} 3 & 2 & 2 \\ 2 & 3 & -2 \end{bmatrix}$.

(2) Suppose $A = U\Sigma V^\top$ is a singular value decomposition with

$$U = \begin{bmatrix} u_1 & u_2 & u_3 \\ u_4 & u_5 & u_6 \\ u_7 & u_8 & u_9 \end{bmatrix}, \quad \Sigma = \begin{bmatrix} 71 & 0 & 0 \\ 0 & 31 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \quad V^\top = \begin{bmatrix} v_1 & v_2 & v_3 \\ v_4 & v_5 & v_6 \\ v_7 & v_8 & v_9 \end{bmatrix}.$$

Find a basis for $\mathrm{Col}(A)$ and a basis for $\mathrm{Nul}(A)$.

(3) Let $\mathbb{D}^2$ be the set of vectors $v \in \mathbb{R}^2$ with $\|v\| = 1$. Suppose A is a 2×2 matrix with $\min\{\|Av\| : v \in \mathbb{D}^2\} = 20$ and $\max\{\|Av\| : v \in \mathbb{D}^2\} = 22$. Assume that $A \begin{bmatrix} 3 \\ 4 \end{bmatrix} = \begin{bmatrix} 0 \\ 100 \end{bmatrix}$. Draw a picture of the region $\{Av : v \in \mathbb{D}^2\}$ in $\mathbb{R}^2$. Then determine all possible values for A.